

The design of GaugeTheory/

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Text in blue was added after the first draft: sums of contents (Nathaneal, 23 September) and invariant reduction (24 September).

1 Purpose

Physlib/ClassicalFieldTheory/GaugeTheory/ is the symbolic framework for the local content of a gauge theory: the algebra in which a Lagrangian is written, the gauge and Lorentz actions on it, and the invariants of a given mass dimension. Fields enter only through their jets at one point, the formal symbols $\partial_s \psi_\alpha$.

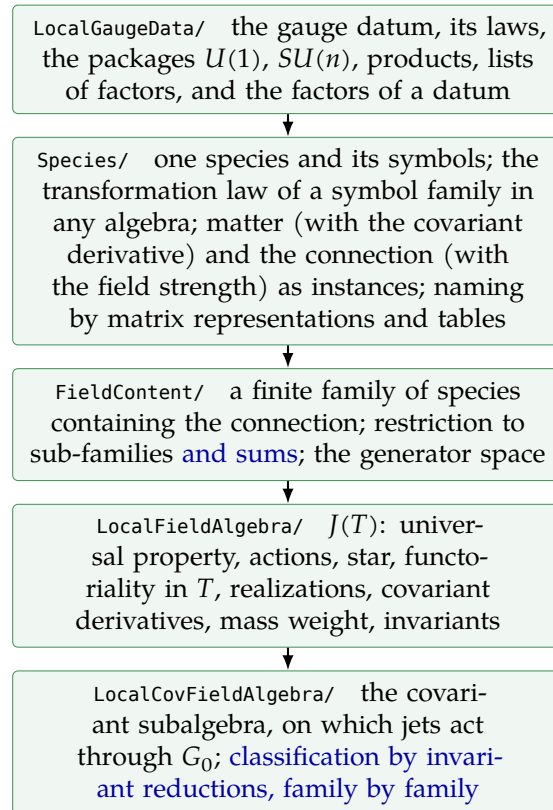
One rule organises the folder: **every layer takes a datum and returns constructions and theorems; a concrete theory contributes only its data**. The Standard Model is a list of gauge factors and a table of fields, and everything else about it is inherited. This note records the intended structure. The present tree differs from it (§6), and the open choices are in §5.

2 The idea

- **Gauge jets.** A gauge transformation is its jet U in a jet group G_J : value eval $U \in G_0$, Lie algebra $\mathfrak{g}_J$, derivatives, adjoint action, Maurer–Cartan form $\omega_\mu(U) = i(\partial_\mu U)U^{-1}$. These maps and their laws are the gauge datum; nothing downstream knows which group it is.
- **Species.** A species is a finite-dimensional V with a Lorentz representation, a parity, a mass weight, and an affine action of G_J on $\text{JetRing} \otimes V$, a linear part ℓ and a translation t with $t(UV) = t(U) + \ell(U)t(V)$. Matter has $t = 0$ and ℓ fibrewise, generated by an infinitesimal action of $\mathfrak{g}$. The connection has $\ell = \text{Ad}$, $t = \omega$.
- **The algebra.** A content T is a finite family of species containing the connection. $J(T)$ is the free graded-commutative algebra on the symbols of all species. Every action is defined on generators and extended by the universal property; the translation part gives the inhomogeneous term. Reality is a star structure. An inclusion of contents $S \subseteq T$ gives an injective equivariant map $J(S) \rightarrow J(T)$ with a retraction: sectors are its images, and the pure gauge theory is the connection-only content. **Contents also add: $F + G$ is the union of the two families glued along the connection, and $J(F + G)$ is the graded tensor product of $J(F)$ and $J(G)$ over the connection-only algebra, graded by which summands a word touches; both actions preserve the grading.**
- **Invariants.** Mass weight is twice the mass dimension (derivative 2, scalar 2, Weyl fermion 3, connection 2); the dimension-four Lagrangian is the gauge and Lorentz invariants of weight at most 8. On the covariant subalgebra jets act through G_0 , so classification is ordinary invariant theory, sector by sector. **It is done one family of terms at a time: a *reduction* `ReducesInvariantsTo` σ V W says that for every σ -stable submodule S , an invariant of $V \sqcup S$**

lies in $W \sqcup S$. Reductions compose, combine over finite joins and transport along maps of index sets, and the family σ is any family of linear maps, so gauge and Lorentz invariance are one invariance for the disjoint union of the two groups. A classification of one family is a reduction of it to the span of its allowed contractions, proved in the quotient by the remainder. In particular, the invariants of an extension $F + G$ are those of F , those of G and those of the mixed part, and extending a classified theory (right-handed neutrinos, SMEFT operators, a new scalar) needs only the mixed invariants.

3 The layers



Imports go down the line only. Each layer exposes one structure as its datum, LocalGaugeData, Species, FieldContent, and proves everything for an arbitrary instance. Two points (now three) decide the shape:

- **One species notion.** Matter fields and the connection are both species, so the jet component space, the transformation law of a symbol family and the realization notion are stated once. The factors of a gauge datum belong to the gauge layer, not to matter.
- **One algebra tower.** The pure gauge theory is J of the connection-only content, and the sectors of $J(T)$ are images of sub-contents. Nothing is built twice and identified afterwards.
- **One classification mechanism.** The relation ReducesInvariantsTo Of Mathematics/InvariantReduction is general mathematics and stays there. The algebra layer supplies what every theory needs to use it: the gauge and Lorentz actions as one family indexed by $G_0 \sqcup SL(2, \mathbb{C})$, the reduction of a factor's invariance to that family, and the stability of symbol ranges. A classification theorem for one index law (a colour trace, an isospin contraction, a Lorentz scalar) is stated as a reduction modulo an arbitrary stable submodule, so it can be applied inside any larger expression and to any extension.

The subsections below describe each layer in detail: its datum, what it derives from that datum, the concrete instances it provides, and the choices behind its shape.

3.1 LocalGaugeData/: the gauge datum

Role. A local Lagrangian sees a gauge transformation $g(x)$ only through its jet at the base point. This layer says, once and for all gauge groups, what structure of those jets the rest of the library may use. Nothing in it mentions matter, a field content or an algebra of symbols; every other layer takes a term of its structure as an argument.

The datum. `LocalGaugeData` $G_0 \ \mathfrak{g} \ G_J \ \mathfrak{g}_J$ has four carriers: the gauge group G_0 , its Lie algebra $\mathfrak{g}$, the group G_J of jets of gauge transformations and the Lie algebra $\mathfrak{g}_J$ of jets of Lie algebra elements. Its maps and laws are exactly those that the transformation laws of fields use:

- evaluation at the base point and inclusion of constants, on the group (`eval`, `ofConstant`) and on the Lie algebra (`evalLie`, `ofConstantLie`), with `eval` $\circ$ `ofConstant` = `id`;
- commuting formal derivatives ∂_μ on $\mathfrak{g}_J$, derivations of the bracket that kill constants, and multiplication by the coordinates x_μ , with $\partial_\mu(x_\nu a) = x_\nu \partial_\mu a + \delta_{\mu\nu} a$;
- the adjoint action of G_J on $\mathfrak{g}_J$ by Lie automorphisms, evaluating to the adjoint action of G_0 on $\mathfrak{g}$;
- the Maurer–Cartan form $\omega_\mu(U) = i(\partial_\mu U)U^{-1}$, zero on constants, with the cocycle law $\omega(UV) = \omega(U) + \text{Ad}_U \omega(V)$, the flatness equation $\partial_\mu \omega_\nu - \partial_\nu \omega_\mu + [\omega_\mu, \omega_\nu] = 0$ and the Leibniz rule $\partial_\mu(\text{Ad}_U x) = \text{Ad}_U \partial_\mu x - [\omega_\mu, \text{Ad}_U x]$.

The Maurer–Cartan form is a field of the structure rather than a construction: the Leibniz rule determines it only up to the centre of $\mathfrak{g}_J$, which is nonzero whenever there is a $U(1)$ factor (`maurerCartan_eq_of_deriv_adjoint`).

What follows from the laws alone. Three developments use nothing but the structure, so they hold for every gauge group at once.

- *Adjoint Taylor coefficients* (`AdjointCoeff`). The coefficient $\partial_x(\text{Ad}_U)|_0$, a linear map on $\mathfrak{g}$, is what the gauge-field law sees of Ad_U after x derivatives. The Taylor–Leibniz theorem says the Taylor data of $\text{Ad}_U Y$ are the antidiagonal convolution of those of Ad_U and of Y ; for matrices it is the Leibniz rule, here it is derived from the single-derivative law by induction. The coefficients are multiplicative up to convolution.
- *The symmetrised Maurer–Cartan form* (`MaurerCartan`). $\bar{\omega}_r = \frac{1}{|r|} \sum_{\mu \in r} \partial_{r-\mu} \omega_\mu$ averages over which direction is carried by the form. Flatness makes every $\partial_s \omega_\mu$ equal to $\bar{\omega}_{\mu::s}$ plus brackets in fewer directions, so the symmetrised base-point data determine all base-point Taylor data of ω .
- *The truncation filtration* (`Truncation`). A jet is trivial to order n when its value is 1 and the Taylor data of ω vanish below order n . These form normal subgroups; the zeroth are the pure jets, and every jet is a pure jet times the constant jet of its value. A jet trivial to order n acts on the connection symbols with at most n derivatives by a pure translation.

The two properties. Two facts true of every honest jet group do not follow from the laws, and are separate classes:

- **Faithful:** a Lie algebra jet is determined by its base-point Taylor data, and a jet with vanishing Maurer–Cartan form is constant;
- **Free** (extends **Faithful**): every family of Taylor data in $\mathfrak{g}$ is realised by a Lie algebra jet, and every $\rho \in \mathfrak{g}_J$ vanishing at the base point is the radial component $\sum_\mu x_\mu \omega_\mu(U)$ of some pure jet.

Free makes the symmetrised Maurer–Cartan data free coordinates on the pure jets. It is what the gauge-boson layer needs to show that gauge invariance forbids symmetrised derivatives of the connection, i.e. that the connection enters invariants only through the field strength.

Concrete packages. All concrete gauge data are built by three constructions.

- **Matrix presentations** (**MatrixJets**). A presentation gives injective maps of the four carriers into $\kappa \times \kappa$ matrices (of $\mathbb{C}$ or of jets) and says that each structure map is the matrix one: entrywise constant coefficient and constants, entrywise derivative and coordinate, conjugation $UaU^\dagger$ for the adjoint actions, $i(\partial_\mu U)U^\dagger$ for ω . From it the laws, **Faithful**, a criterion for **Free** and the canonical SU -type factor are proved once, from matrix identities over the jet ring. $\mathfrak{su}_n$ (special unitary matrices of jets, traceless hermitian Lie algebra) and $\mathfrak{u}_1$ (unitary scalars read as 1×1 matrices) are instances; a $U(n)$ or $SO(n)$ package would only supply its structure maps.
- **Products** (**Prod**). Every map and law acts componentwise; **Faithful** and **Free** pass to products.
- **Lists of factors** (**OfFactors**). A model names its gauge group by a list such as `[.SU 3, .SU 2, .U1]`; **ofFactors** is the product of the packages in that order, and it is free. Its carriers are written by recursion so that they unfold to the literal product $\text{JetSU } 3 \times \text{JetSU } 2 \times \text{JetU1}$.

Factors. A *factor* of a datum is a projection onto $U(1)$ or $SU(n)$ that is compatible with the structure: a unitary (matrix of) jets $u(U)$ for each gauge jet, with matching components φ, φ_J on the Lie algebras, related by $\varphi_J(\omega_\mu U) = i(\partial_\mu u)u^{-1}$ and by the adjoint action (**U1Factor**, **SUFactor**). A gauge group presented by its factors is a list **Factors jets**. **Factors** are the interface to the next layer: the charges of a species are given factor by factor, and the representation they name (a charge twist, a fundamental) is built there. **ofFactors** comes with its canonical factors.

Files.

Basic	the structure, first consequences, iterated derivative, Faithful
AdjointCoeff	adjoint Taylor coefficients, the Taylor–Leibniz theorem
MaurerCartan	the symmetrised form; symmetrised data determine ω
Truncation	the filtration, pure jets, radial component, Free
Factor	U1Factor , SUFactor , Factor , Factors
MatrixJets	presentations by matrices; laws, faithfulness, freeness, factor
U1, SU/Algebra, SU/Basic	the packages u1 , $\mathfrak{su}_n$ as instances
Prod, OfFactors	products and lists of factors

Choices. A datum is passed as an argument, never inferred: the carriers do not determine it (a truncated jet group has the same carriers), so there is nothing canonical for instance search to find. The layer imports nothing above it, so that the gauge data of a model can be read without the rest of the library.

4 What a model sees

A model is a card and nothing else:

```
abbrev gauge : List FactorSpec := [.SU 3, .SU 2, .U1]
abbrev leptonDoublet : SpeciesData factors := (.L, .singlet, .fund, -3)
...
noncomputable def content : FieldContent gaugeData := table.toFieldContent
```

Species of the model are abbreviations of the value spaces the card produces, and theorems are stated in the card and the generic vocabulary alone, for example that the gauge and Lorentz invariants of content of weight at most 4 are spanned by 1 and H^+H .

5 Questions for discussion

1. **One species structure or two.** The affine law costs a translation field that is zero for all matter. Is the uniformity worth it, or should `MatterField` and the connection stay separate with the symbol-family law stated twice?
2. **Real or complex tower.** The connection's algebra is real today; the single tower is complex with a star. Does anything downstream need the real algebra as an object?
3. **The connection in the content.** Explicit member of the family (restriction becomes ordinary) or implicit through the gauge datum (as today)? [Explicit forces a sum \$F + G\$ to be glued along the connection; implicit allows a plain disjoint union.](#)
4. **Sectors by functoriality.** Restriction along $S \subseteq T$ fixes how species are indexed. Is that acceptable to the existing sector proofs? [Images of sub-contents give the "at most \$S\$ " subalgebras, but those proofs use the "exactly \$S\$ " pieces, and a sum has a mixed part in neither image. Should the grading by sub-families be part of this layer?](#)
5. **Names and order of work.** `Species`, `FieldContent`, `content` against the current names; and which of the steps below, if any, should precede the remaining Standard Model migration?

6 Distance from the current tree

current	intended
<code>LocalGaugeData/</code>	the same, plus the factor types now in <code>MatterField/MatrixRep/</code>
<code>MatterField/; GaugeBoson/Basic, GaugeBoson/Realization/</code>	<code>Species/</code>
<code>GaugeBoson/LocalGaugeFieldAlgebra/, GaugeBoson/LocalGaugeCovFieldAlgebra/</code>	the two algebra layers at the connection-only content
<code>GaugeFieldData/</code>	<code>FieldContent/</code> , with the connection explicit
<code>LocalFieldAlgebra/, LocalCovFieldAlgebra/</code>	the same, sectors derived from functoriality
<code>Mathematics/InvariantReduction;</code>	the same; the combined family, the factor reductions
<code>StandardModel/InvariantReduction</code>	and the range stabilities move into <code>LocalCovFieldAlgebra/</code>, stated for <code>ofFactors</code>

In the order they should be removed:

1. `LocalGaugeData/` imports `matter` (the factor types, and `TransformsIn`, `InfinitesimalAction` importing the covariant derivative). Move the factor types down and the two files into the species layer.

2. Matter and the connection have differently shaped APIs, `TransformsIn` for matter and `GaugeAlgebraRealization` for the connection. Introduce the species structure with both as instances.
3. Sectors are built by hand and the gauge sector identified with a separate algebra (four `GaugeSector` files). Prove functoriality and derive them.
4. The connection has its own real algebra tower of about 3300 lines. Replace it by the connection-only content, last, because the Standard Model gauge-sector classification is built on it.

Steps 1 and 2 are cleanup; 3 and 4 change shared conventions and need their owners.

7 Progress

Kept up to date as the layers are reorganised. Each entry says what was changed and what the layer now is, so that a reviewer can check the commits against the intent above.

`LocalGaugeData/` (24 September 2026, 9 commits for PR #1415)

Changed. The layer imported matter in two ways; both are gone. The factor types `U1Factor`, `SUFactor`, `Factor`, `Factors` moved down from the matrix-representation and table files into a new `Factor.lean`, since their fields mention only gauge jets, the gauge algebra and the jet ring. `TransformsIn` and `InfinitesimalAction`, which are statements about a representation on $\text{JetRing} \otimes V$, moved up to `MatterField/`. The carriers of a list of factors were renamed from `Factors.G` to `OfFactors.G`, so that `Factors` names only the type. The duplication between `U1.lean` and `SU/Basic.lean` was removed by a new `MatrixJets.lean`: a presentation of gauge jets by matrices of jets (injective maps of the four carriers into $\kappa \times \kappa$ matrices, and what each structure map is in matrices), from which the laws of `LocalGaugeData`, faithfulness and the canonical `su`-type factor are proved once; the matrix identities over the jet ring it needs went to `Relativity/JetRing/Matrix`. `su n` and `u1` are now instances, `u1` reading a scalar as a 1×1 matrix. Eight lemmas referenced nowhere, a stale TODO and a helper definition were removed, and the TODO on the recursive carriers was replaced by the reason for the recursion.

Now. Eleven files, about 3 500 lines, importing nothing outside the layer. `Basic` (the structure, its laws, the iterated derivative, `Faithful`); `AdjointCoeff` (the Taylor–Leibniz theorem); `MaurerCartan` (the symmetrised form and what determines the form); `Truncation` (the filtration, pure jets, `Free`); `Factor` (the factor types); `MatrixJets` (the presentation by matrices); `SU/Algebra`, `SU/Basic`, `U1` (the concrete packages as instances); `Prod` (products and lifting factors); `OfFactors` (a list of factor symbols, its carriers and canonical factors).

Follow-up (Joseph, 25 September). Freeness is now also proved once for a matrix presentation: `MatrixJets.free` reduces `Free` to three conditions on the carriers (Lie algebra jets contain every matrix with Taylor data in $\mathfrak{g}$, they are hermitian, and the gauge jets contain the unitary Euler transport of every Lie algebra jet), using the Euler-transport results on matrices of jets. The Standard Model freeness proof becomes an instance and shrank by about 400 lines. The layer is now about 3 600 lines, still importing nothing outside itself.

Not done. `U1.lean` and `SU/Basic.lean` still each define their structure maps by hand (about 150 lines each); a further step would define the carriers themselves as subtypes of matrices and take the maps from `MatrixJets`, but that changes the carrier types the Standard Model files unfold to, so it waits for the hand-built Standard Model code to go.